Problem Sheet #1

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Problem 1:
If A and B are sets, show that $A \subseteq B$ if and only if $A \cap B = A$.

Solution:
We proceed by mutual implication.

Forward Direction ($\Longrightarrow$): Assume $A \subseteq B$. We must show $A \cap B = A$. First, let $x \in A \cap B$. By the definition of intersection, $x \in A$ and $x \in B$. Thus, $x \in A$, which implies $A \cap B \subseteq A$. Second, let $x \in A$. Since $A \subseteq B$, it follows that $x \in B$. Because $x \in A$ and $x \in B$, we have $x \in A \cap B$. Thus, $A \subseteq A \cap B$. By mutual inclusion, $A \cap B = A$.

Reverse Direction ($\Longleftarrow$): Assume $A \cap B = A$. We know by that $A \cap B \subseteq B$, using our assumption we conclude that $A \subseteq B$

Q.E.D.

Problem 2:
If A is a set and $\mathcal{B} = \{B_i : i \in I\}$ is any family of sets, show that:

$$A \cap \bigcup_{i \in I} B_i = \bigcup_{i \in I} (A \cap B_i) \text{ and } A \cup \bigcap_{i \in I} B_i = \bigcap_{i \in I} (A \cup B_i) \tag{1}$$

Solution:
First part.

Forward Direction ($\Longrightarrow$): Let $x \in \left(A \cap \bigcup_{i \in I} B_i\right)$.
By definition of intersection we have $x \in A \wedge \exists j \in I$ such that $x \in B_j$, so we have that $x \in A \cap B_j$, if this element is in the intersection of a sole set of the family, is also true that is in the union of all sets in the family, therefore $x \in \bigcup_{i \in I} (A \cap B_i)$

Reverse Direction ($\Longleftarrow$): Let $x \in \bigcup_{i \in I} (A \cap B_i)$
We know that $\exists j \in I$ such that $x \in B_j \cap A$, this is $x \in A \wedge x \in B_j$, because x is in any set of the family of sets, is also true that is in the union of the family, therefore $x \in \bigcup_{i \in I} B_i \wedge x \in A$, so we have that $x \in A \cap \bigcup_{i \in I} B_i$

Second part.

Forward Direction ($\Longrightarrow$): Let $x \in A \cup \bigcap_{i \in I} B_i$
So $x \in A \ \vee \ \forall j \in I, \ x \in B_j$ this is $x \in A \cup B_j$, for this we know that $\forall j \in I, \ x \in A \cup B_j$, this is $\bigcap_{i \in I} (A \cup B_i)$

Reverse Direction ($\Longleftarrow$): Let $x \in \bigcap_{i\in I}(A \cup B_i)$
so $\forall j \in I,\ x \in A \cup B_j$, so if $\forall j \in I,\ x \in A \vee x \in B_j$,by Law of excluded middle we divide this in two cases:

1. $x \notin A$
   - If this is the case, then $x \in B_j, \forall j \in I$ must be true, but this is trivially $x \in A \cup \bigcap_{i\in I} B_i$
2. $x \in A$
   - This is trivially $x \in A \vee \forall j \in I,\ x \in B_j$ therefore $x \in A \cup \bigcap_{i\in I} B_i$

Q.E.D.

Problem 3:
Let B be any set. Show that $(A \cap B) \cup C = A \cap (B \cup C) \iff C \subseteq A$.

Solution:
Forward Direction ($\Longrightarrow$): Let $(A \cap B) \cup C = A \cap (B \cup C)$, suppose $x \in C$, we need to prove that this will also hold for $x \in A$.

If $x \in C$ then by $A \cap (B \cup C)$ we have that $x \in A \wedge (x \in B \vee x \in C)$, therefore $x \in A$, thus $x \in C \Rightarrow x \in A$ that translates to $C \subseteq A$

Reverse Direction ($\Longleftarrow$): Let $C \subseteq A$
So if $x \in C$ then $x \in A$, we know that $C \cap A \subseteq A$